

	Setup Time	<i>clk</i> -to- <i>Q</i> Delay	<i>D</i> -to- <i>Q</i> Delay	Contamination Delay	Hold Time
Flip-Flops	65 ps	50 ps	n/a	35 ps	30 ps
Latches	25 ps	50 ps	40 ps	35 ps	30 ps

Question 1

For each of the following sequencing styles, determine the maximum logic propagation delay available within a 500 ps clock cycle. Assume there is zero clock skew.

- (a) Flip-flops
- (b) Two-phase transparent latches

Question 2

For each of the following sequencing styles, determine the maximum logic propagation delay available within a 500 ps clock cycle if the clock skew between any two elements can be up to 50 ps.

- (a) Flip-flops
- (b) Two-phase transparent latches

Question 3

For each of the following sequencing styles, determine the minimum logic contamination delay in each clock cycle (or half-cycle, for two-phase latches). Assume there is zero clock skew.

- (a) Flip-flops
- (b) Two-phase transparent latches with 50% duty cycle clocks
- (c) Two-phase transparent latches with 60 ps of nonoverlap between phases

Question 4

For each of the following sequencing styles, determine the minimum logic contamination delay in each clock cycle (or half-cycle, for two-phase latches) if the clock skew between any two elements can be up to 50 ps.

- (a) Flip-flops
- (b) Two-phase transparent latches with 50% duty cycle clocks
- (c) Two-phase transparent latches with 60 ps of nonoverlap between phases

Question 5

Consider a flip-flop built from a pair of transparent latches using nonoverlapping clocks. Express the setup time, hold time, and clock-to-Q delay of the flip-flop in terms of the latch timing parameters and $t_{\text{nonoverlap}}$, relative to the rising edge of ϕ_1 .

Question 6

Determine the minimum clock period at which the circuit in Figure below will operate correctly for each of the following logic delays. Assume there is zero clock skew and that the latch delays are accounted for in the propagation delay

- (a) $\Delta_1 = 300$ ps; $\Delta_2 = 400$ ps; $\Delta_3 = 200$ ps; $\Delta_4 = 350$ ps
 (b) $\Delta_1 = 300$ ps; $\Delta_2 = 400$ ps; $\Delta_3 = 400$ ps; $\Delta_4 = 550$ ps
 (c) $\Delta_1 = 300$ ps; $\Delta_2 = 900$ ps; $\Delta_3 = 200$ ps; $\Delta_4 = 350$ ps

